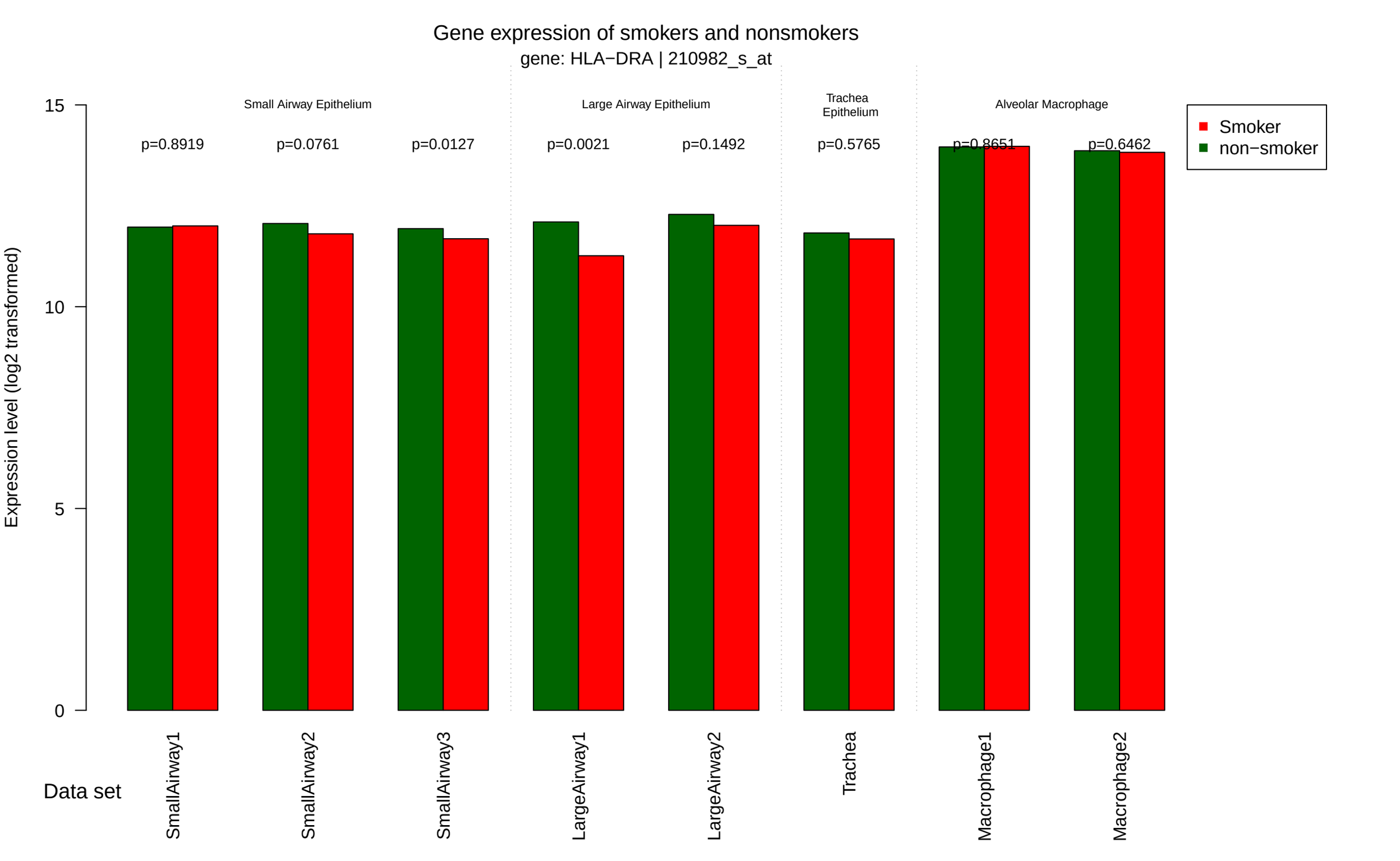
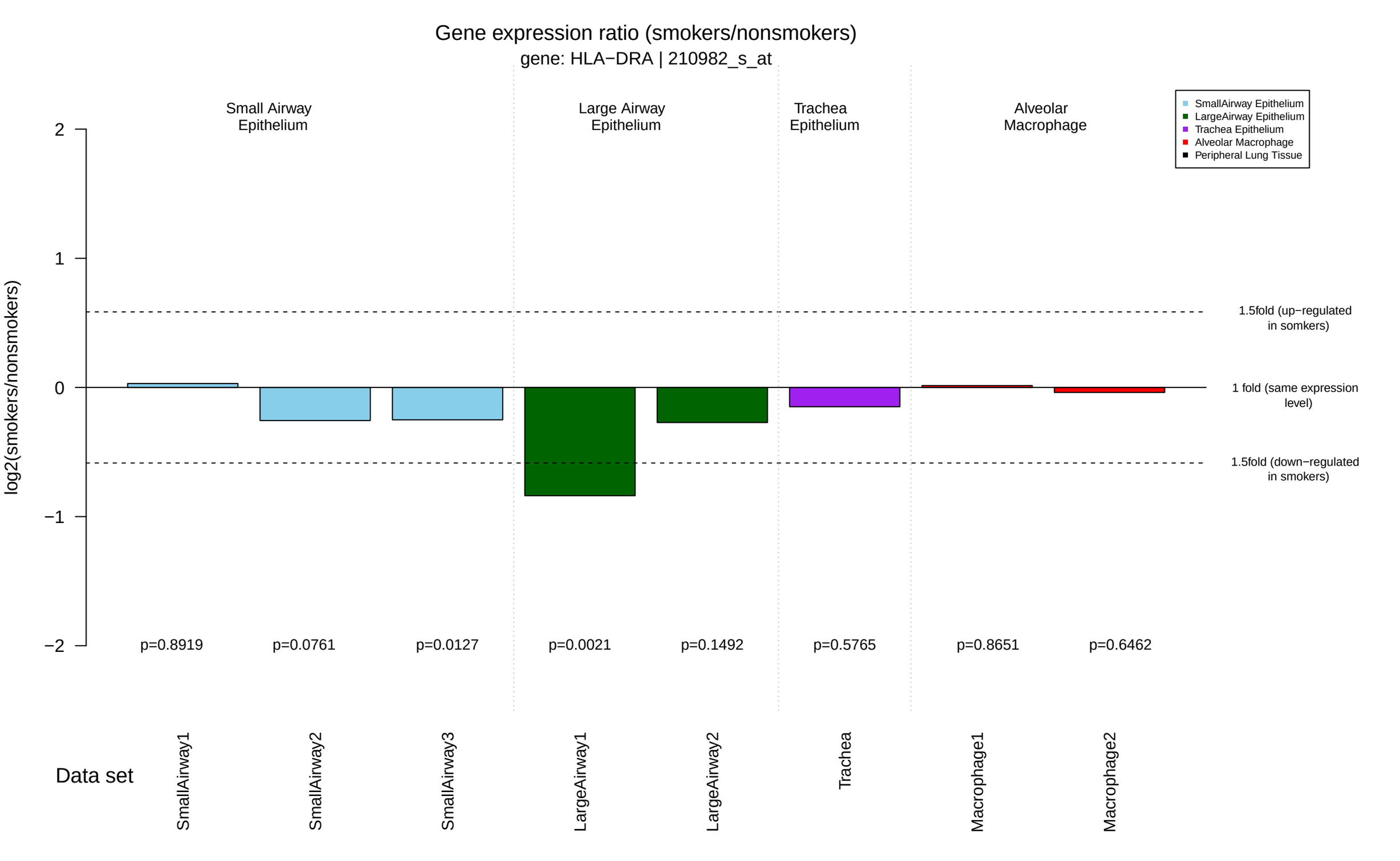
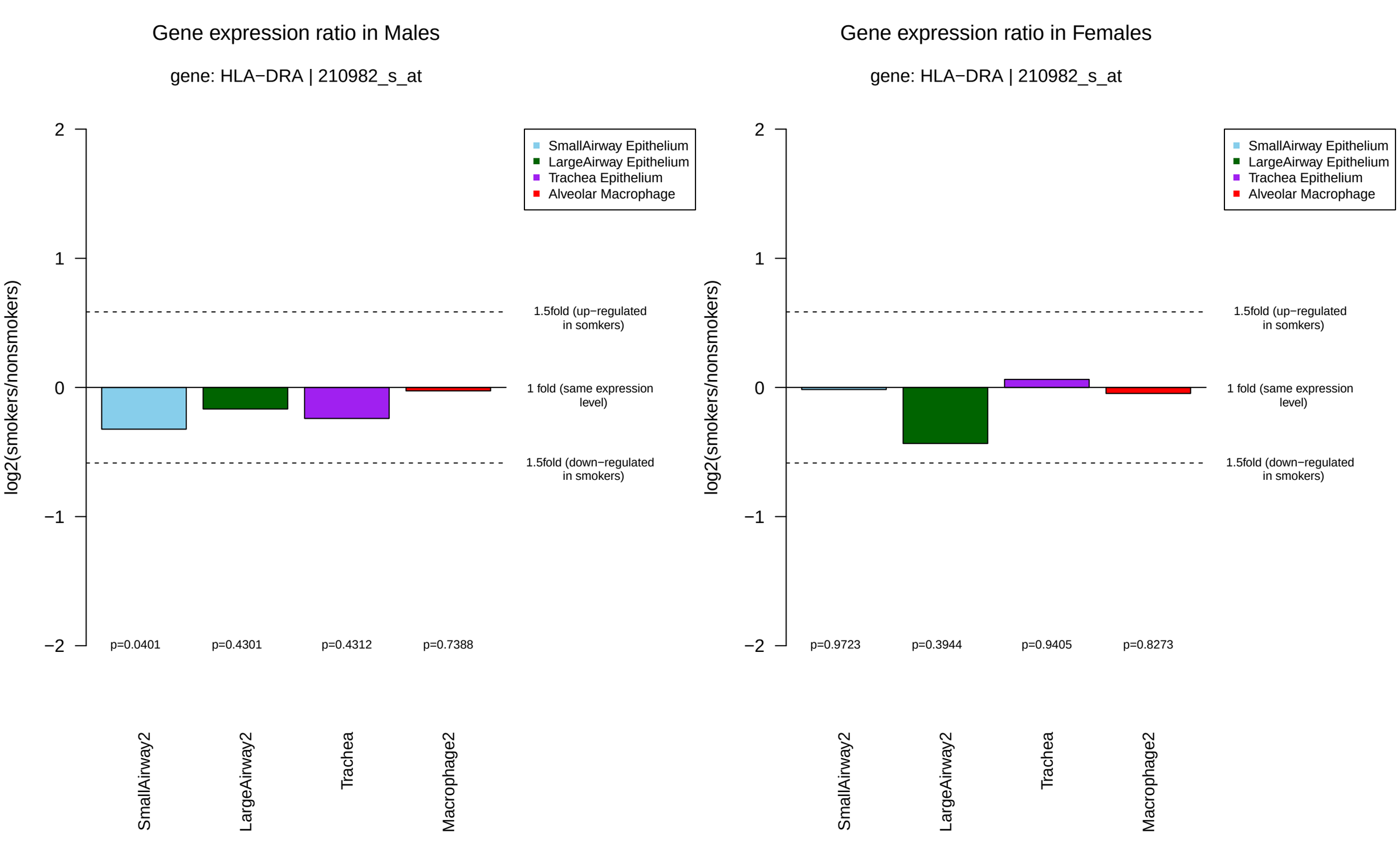
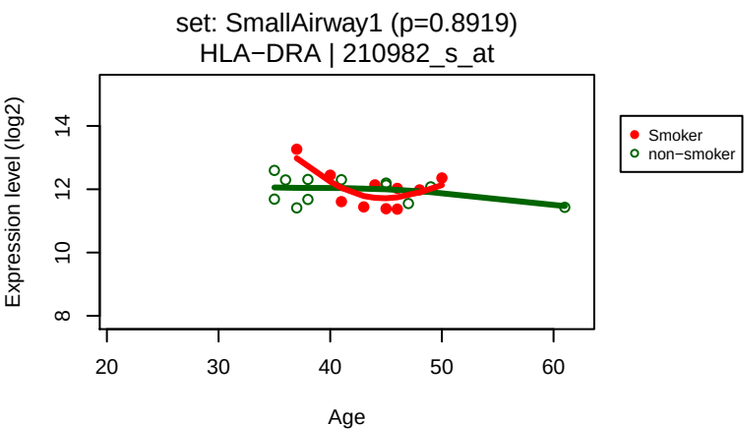
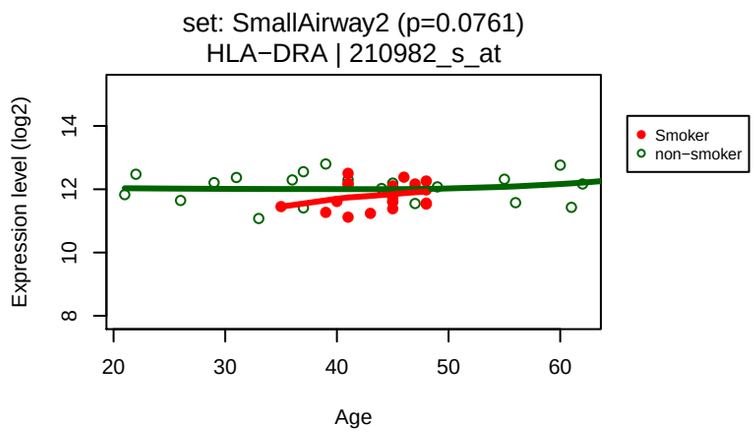
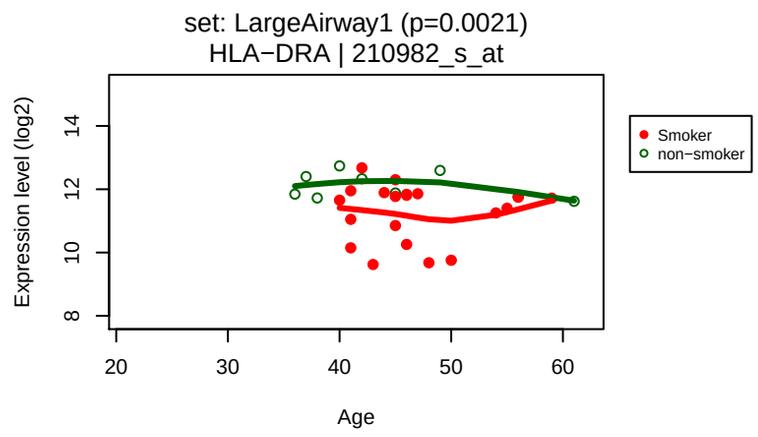

Gene expression of smokers and nonsmokers
gene: HLA−DRA | 210982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8919
p=0.0761
p=0.0127
p=0.0021
p=0.1492
p=0.5765
p=0.8651
p=0.6462
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

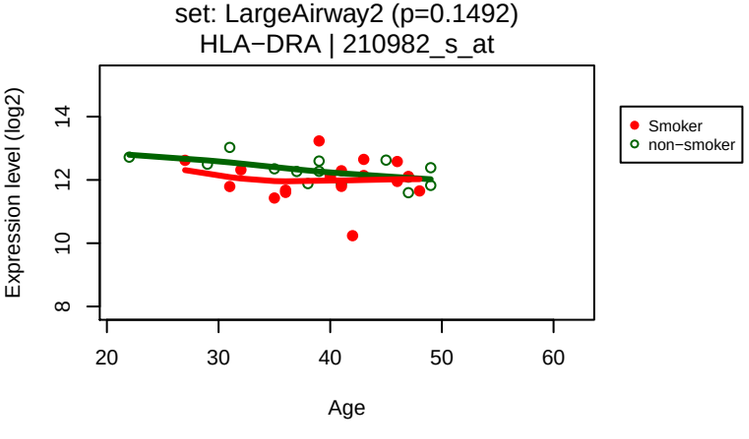

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRA | 210982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8919
p=0.0761
p=0.0127
p=0.0021
p=0.1492
p=0.5765
p=0.8651
p=0.6462
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

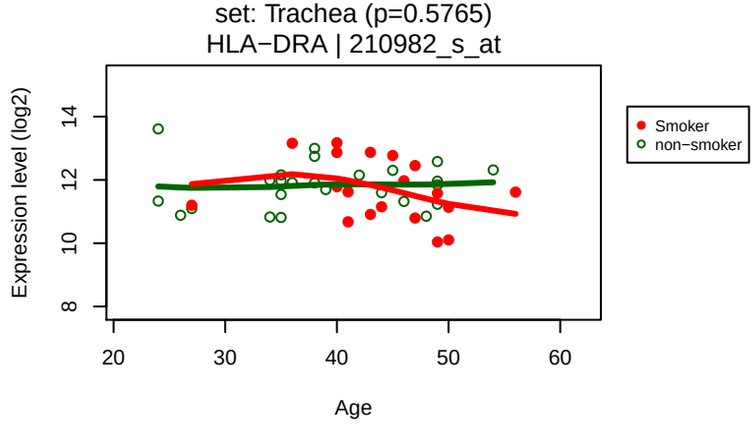

Gene expression ratio in Males
gene: HLA−DRA | 210982_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0401
p=0.4301
p=0.4312
p=0.7388
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DRA | 210982_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9723
p=0.3944
p=0.9405
p=0.8273
SmallAirway2
LargeAirway2
Trachea
Macrophage2

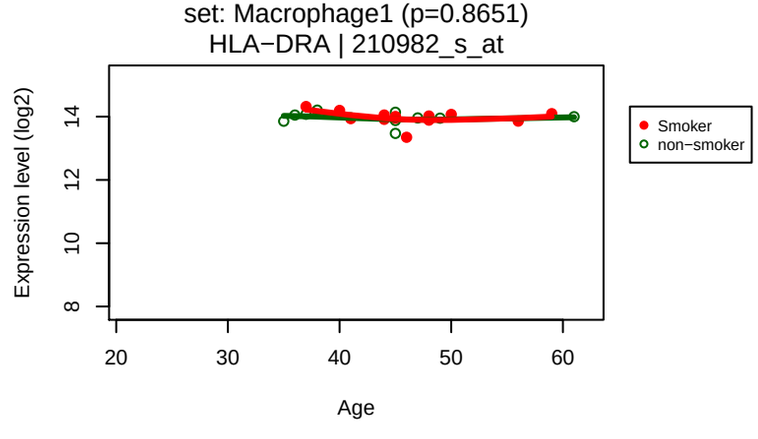

set: SmallAirway1 (p=0.8919)
HLA−DRA | 210982_s_at
Expression level (log2)
Age
Smoker
non−smoker

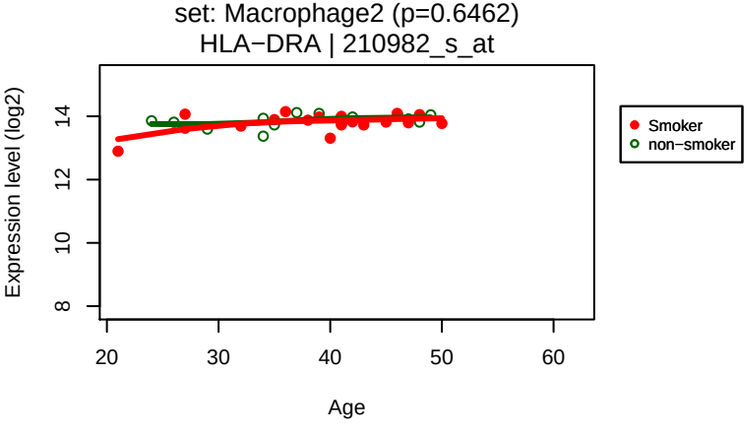

set: SmallAirway2 (p=0.0761)
HLA−DRA | 210982_s_at
Expression level (log2)
Age
Smoker
non−smoker

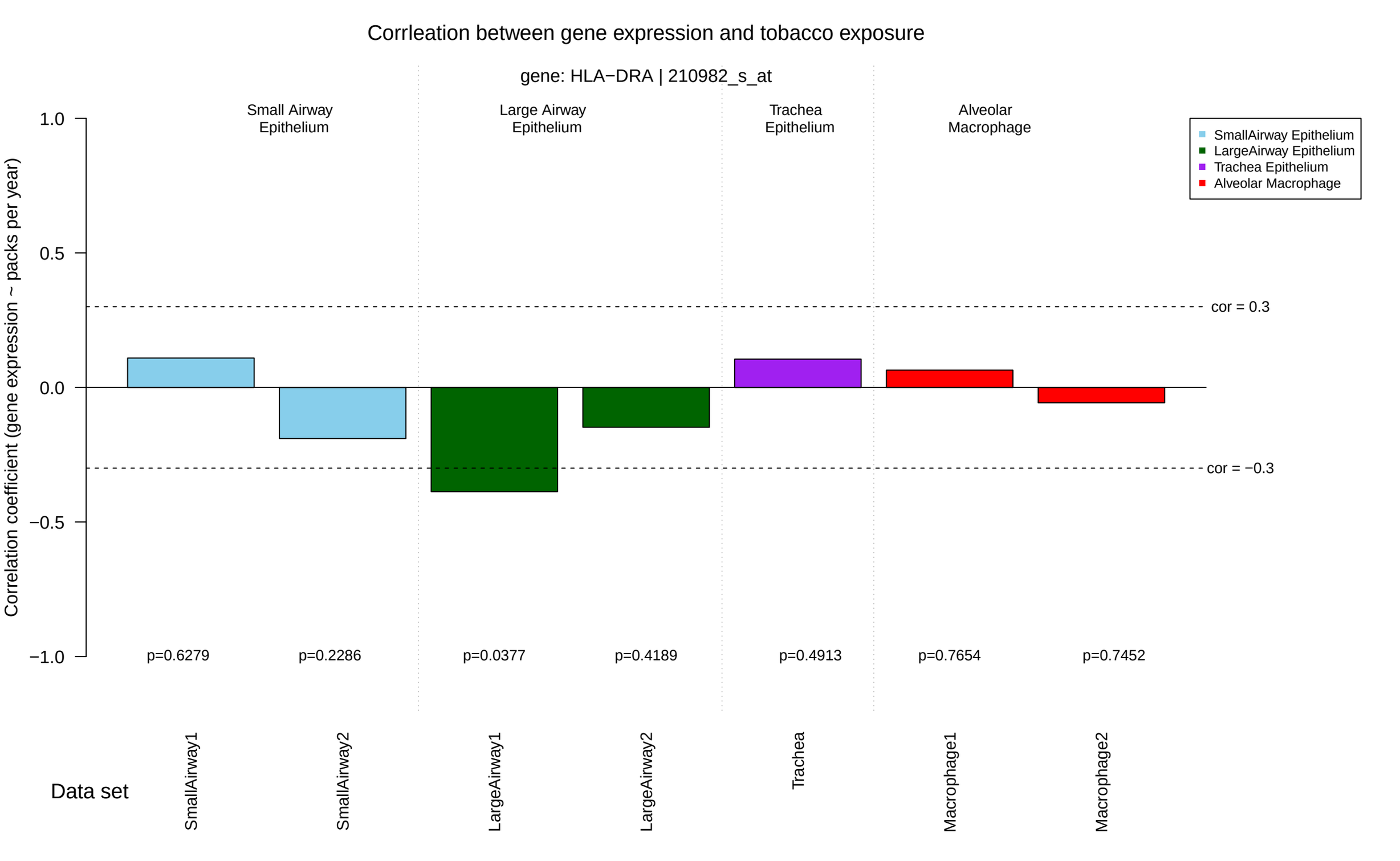

set: LargeAirway1 (p=0.0021)
HLA−DRA | 210982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1492)
HLA−DRA | 210982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5765)
HLA−DRA | 210982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8651)
HLA−DRA | 210982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6462)
HLA−DRA | 210982_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRA | 210982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6279
p=0.2286
p=0.0377
p=0.4189
p=0.4913
p=0.7654
p=0.7452
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2